# Prabhavit Gupta



Freelance AI Engineer: LLMs, RAG Architectures & AI Agent Workflows

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## Summary

Freelance AI Engineer with 3 years of hands-on production experience architecting, deploying, and operating enterprise AI solutions, RAG retrieval pipelines, and autonomous AI agents. Proven track record building and shipping production AI systems utilizing modern LLM platforms (including Google Cloud / Vertex AI and OpenAI/Anthropic APIs), vector search datastores, and tool-augmented agent workflows. Experienced translating complex enterprise requirements into scalable, reliable Python and TypeScript microservices with rigorous evaluation harnesses and human-in-the-loop controls. Holder of an EU Long-Term Residence Permit (permanent unrestricted EU work authorization across all EU member states, no sponsorship required); based in Venice, Italy and available immediately for European remote freelance contracts.

## Technical Expertise

- **Generative AI, LLMs & Agent Systems:** Multi-agent workflows, autonomous tool calling, LangChain, prompt engineering, structured JSON extraction, human-in-the-loop approvals, transcript evaluation harnesses.
- **AI Platforms & Cloud Ecosystems:** Google Cloud (Vertex AI), OpenAI API, Anthropic Claude, Docker containerization, CI/CD pipelines, Git version control, Linux server management.
- **RAG Architectures & Vector Retrieval:** Retrieval-Augmented Generation, chunking algorithms, hybrid search (dense + BM25), PostgreSQL with `pgvector`, metadata filtering, context reranking, citation grounding.
- **Software Development & Backend Systems:** Python (FastAPI, asyncio, pandas), TypeScript, Node.js, RESTful microservices, webhook orchestration, PostgreSQL schema modeling, Redis asynchronous task queues.
- **Languages & Work Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (unrestricted EU work authorization across all EU member states, no sponsorship required).

## Experience

**FlowSpace.co** **Mar 2024 - Aug 2026**
*Lead Developer & AI Systems Architect* *London, UK (remote)*

- Architected and deployed production customer-facing AI agents and retrieval pipelines for an enterprise platform serving 4,500+ active business users, integrating platform catalogs with external APIs.
- Engineered automated document parsing and ingestion pipelines using LLMs, converting unstructured partner data into validated JSON schemas and expanding live platform inventory.
- Designed multi-step agent reasoning chains with strict tool execution constraints, schema validation, and human-in-the-loop approval checkpoints, ensuring zero ungrounded actions in customer-facing flows.
- Built automated evaluation harnesses evaluating retrieval precision and response accuracy against real user conversation transcripts, reducing edge-case hallucination rates.
- Maintained resilient TypeScript/Node.js backend services and PostgreSQL data layers, utilizing asynchronous Redis queues to handle high-throughput query spikes.
- Implemented automated integration and end-to-end regression test suites with Playwright, verifying API contracts and preventing system regressions across continuous deployment cycles.

**Urlink** **Apr 2025 - Jul 2025**
*Project Manager & Solutions Developer (Freelance)* *Barcelona, Spain (remote)*

- Led technical discovery, solution architecture, and client-facing delivery of customized web solutions and automated data processing pipelines for European client platforms.
- Built automated data ingestion and transformation services connecting third-party partner APIs to PostgreSQL databases, maintaining strict data integrity and schema validation.
- Shipped interactive prototypes and API test harnesses to validate system boundaries and data flows prior to sprint execution, accelerating delivery timelines.

**Independent Technical Consultant** **Aug 2023 - Feb 2024**
*AI Automation & Integration Developer* *Venice, Italy*

- Built and deployed custom LLM integrations, document retrieval prototypes, and webhook automation workflows for European enterprise clients using Python, TypeScript, and n8n.

- Replaced brittle manual document review workflows with automated LLM-assisted classification and extraction pipelines featuring comprehensive error handling and fallback routing.
- Authored technical architecture specifications, API documentation, and deployment guides to support client engineering teams in operating self-hosted workflows.

## Education

**Ca' Foscari University of Venice** **2020 - 2024**
*BA in Philosophy, International Studies and Economics* *Venice, Italy*

- Rigorous training in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.